

Shi Hao Ng

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EDUCATION

Imperial College London

MEng Computing (Artificial Intelligence & Machine Learning)

London, UK

Sep. 2025 – Present

Marlborough College Malaysia

IB Bilingual Diploma: 42 points (777 in HL subjects)

Malaysia

Graduated May 2025

- Extended Essay (Computer Science): Grade A - Pre-training GPT-2 from scratch with modified LayerNorm
- IGCSE: 10A*, 2A. Best in Asia for CS, Best in Malaysia for Design Technology, Physics and Chemistry

EXPERIENCE

Re-Founder & Co-President

Imperial College Drone Society

2025 – Present

London, UK

- Re-established the Imperial College Drone Society, secured 3k+ GBP in funding from SE3.ai, 80+ members
- (In-Progress) Leading team to development of building 400km/h interceptor drone with custom-built gimbal and Raspberry Pi-based computer vision for last-mile tracking
- Organized drone flying lessons (simulation-based) and drone-building workshops

Engineering Intern

Nosco Asia (S) Pte Ltd

2025

Taichung, Taiwan

- Gained hands-on experience in glass manufacturing processes and industrial fabrication techniques
- Developed practical skills in welding and high-temperature materials handling

Full-Stack Developer (Client Work)

Nosco Asia (S) Pte Ltd

2024

Singapore

- Developed workforce management application using ReactJS, Firebase, and Google Cloud Platform
- Built responsive UI with Tailwind CSS and implemented backend services with Node.js and TypeScript
- Deployed scalable enterprise solution integrating real-time data synchronization and user authentication

Founder & Director

Computational Engineering Society

2024 – 2025

Marlborough College Malaysia

- Founded and directed society with 50+ active members focused on computational engineering applications
- Organized workshops on programming, AI, CAD design, and engineering simulation software

PROJECTS

RLAIF Agentic Legal LLM Development | Python, PyTorch, Qwen3-14B, RLAIF | [GitHub](#)

2025

- Led team of 5 at IBM Datathon Z as Team Leader; built full agentic RLAIF pipeline and working training code in 24 hours, going from knowing only theory about RLAIF/GRPO/RAG to a fully working system by end of event
- Fine-tuned Qwen3-14B-Instruct via GRPO with Gemini 2.5 Flash as LLM-as-judge, using LoRA on H100 SXM5
- Generated synthetic QA datasets from FOSS law cases via Gemini 2.5 Pro; built RAG database with tool-use system mimicking real-world lawyer case-retrieval workflows for affordable legal assistance

GPT-2 Pre-training Research | C/CUDA, PyTorch, HPC, Nvidia H100 | [GitHub](#) | [Paper](#)

2024 – 2025

- Pre-trained GPT-2 transformer model from scratch on Nvidia H100 HPC nodes. Experimented with modified Layer Normalization techniques (Pre-LN, RMSNorm, PowerNorm) implemented from scratch in PyTorch
- Implemented distributed training pipeline and monitored convergence metrics across multiple GPU nodes
- Achieved Grade A on IB Extended Essay demonstrating deep understanding of transformer architecture and training dynamics

Google Edutech Asia Challenge - Champion | YOLO-v11, GCP, Firebase | [GitHub](#)

2024

- Served as ML Engineer for drone-based waste detection system winning Champion of Asia award
- Fine-tuned YOLO-v11 CNN model on custom Roboflow trash datasets for real-time object detection
- Created geospatial heatmap analytics visualizing waste concentration patterns for policy stakeholders

- Custom FPV Racing Drone** | *Blender, Betaflight, PET-CF, DJI O4 Pro, Mamba FC* | [LinkedIn Post](#) 2025
- Designed and fabricated 610g racing drone achieving 240 km/h peak speed with custom frame designed in Blender
 - 3D printed frames in PET-CF (carbon fiber reinforced) for optimal strength-to-weight ratio
 - Sourced electronics from Shenzhen including Mamba flight controller, LiPo batteries, and DJI O4 Pro air unit
 - Programmed flight controller via Betaflight CLI/GUI with custom PID tuning for responsive control
- Augmented Railgun Development** | *LS-Dyna, CAD, High-Voltage Electronics* | [Website](#) 2024 – 2025
- Designed and simulated electromagnetic railgun using LS-Dyna FEA software for magnetic field optimization
 - Fabricated acrylic frame via laser cutting and assembled complex high-voltage electrical system
 - Built 450V 10,000µF capacitor bank with hand-coiled ZVS transformer for DC regulation
 - Developed expertise in high-voltage safety protocols and power electronics for nuclear fusion research applications
- F1 in Schools Competition** | *Fusion 360, Ansys Fluent, Mastercam, SolidWorks* | [Website](#) | [Team Site](#) 2023 – 2024
- Chief Engineer and Team Leader for Anduril Racing; sank 200-300 hours designing F1 scale car under strict competition regulations, collaborating with University of Southampton Malaysia for FEA guidance
 - Conducted aerodynamic analysis using Ansys Fluent CFD, optimized car geometry through iterative FEA simulations, then led full pipeline from CAD (Fusion 360/SolidWorks) to CNC machining (Mastercam)
- Custom Game Engine** | *C/C++, OpenGL, Low-Level Programming* | [GitHub](#) 2022 – 2023
- Built complete game engine from scratch in C/C++ implementing custom rendering pipeline with OpenGL
 - Developed low-level memory management system with custom allocators for optimal performance
 - Implemented entity-component system architecture for flexible game object management
 - Created physics engine with collision detection, rigid body dynamics, and spatial partitioning
- 3D Rendering & Graphic Design** | *Blender, Adobe Photoshop and Illustrator* | [MIT Maker Portfolio](#) 2019 – 2021
- Created photorealistic 3D car renders in Blender with advanced lighting, materials, and composition techniques

IN-PROGRESS PROJECTS (Expected Completion: June 2026)

- New Dejima – Autonomous AI Agent Revenue System** | *OpenClaw* | [GitHub](#) | [LinkedIn Post](#) 2026 – Present
- Built off OpenClaw. Built autonomous pipeline where AI agents generate Android apps from natural language, compile, deploy to real hardware, run vision-based Quality Assurance, and self-debug – with no human-in-the-loop
 - Built autonomous marketing engine (ElevenLabs + Veo 3.1 + Lyria) publishing AI-generated videos to YouTube Shorts, achieving 1,200+ views in 30 hours
 - Developed real-time economic tracking layer via Paid.ai measuring revenue-per-token vs cost-per-token; targeting \$10K USD ARR by end of 2026
- First Order Robotics (RoboCup)** | *C++, MPC, Isaac Sim, RL* | [GitHub](#) 2025 – Present
- AI Engineer for Imperial College's RoboCup team developing autonomous football-playing robots
 - Implemented Model Predictive Control (MPC) in C++ for real-time robot trajectory planning and motion control
 - Developing AlphaGo-style self-play RL pipeline using Isaac Sim to replace heuristic-based strategy code
- Interceptor Drone** | *FPV, Raspberry Pi, YOLO, Computer Vision* 2026 – Present
- Leading development of 400km/h interceptor drone with custom gimbal for Imperial College Drone Society
 - Integrating Raspberry Pi with onboard computer vision for autonomous last-mile target tracking
- NVIDIA Isaac Sim - Sim-to-Real Robotics** | *Isaac Sim, RL, SO-101 Arm* | [GitHub](#) | [LinkedIn](#) 2026 – Present
- Training robotic arm (URE5 and SO-101) via reinforcement learning in Isaac Sim with local 5090, after additionally validating policies in MuJuCo and Gazebo, the policies are transferred to real hardware via sim-to-real transfer
 - 3D-printed and assembled SO-101 robotic arm in personal makerspace, will modify SO-101 to improve dexterity
 - Goals: train arm to assemble complex Lego sets (e.g. Millennium Falcon) and handle cable management for data centers server-rack assembly – if a robot can handle this dexterity, it can handle most low-cost manual labour tasks, with significant implications for manufacturing automation

TECHNICAL SKILLS

Languages: Python, C/C++, C/CUDA, Java, Kotlin, JavaScript/TypeScript, Haskell, HTML/CSS
AI/ML Frameworks: PyTorch, TensorFlow, YOLO-v11, HF Transformers, RLAIF, LoRA, GRPO, OpenClaw
Web Development: ReactJS, Node.js, Firebase, Tailwind CSS
Cloud & DevOps: Google Cloud Platform, Git, API Development, HPC Cluster Management, Docker, Vast.ai, AWS
CAD/FEA Simulation: Autodesk Fusion 360, Blender, SolidWorks, Mastercam, Ansys Fluent, Ansys EM, LS-Dyna
Hardware/Robotics: Betaflight, Circuit Design, High-Voltage Systems, 3D Printing, CNC Machining, Soldering
Robotics/Simulation (In-Progress): NVIDIA Isaac Sim, NVIDIA Isaac Lab, MuJuCo, Gazebo, OpenUSD, Model Predictive Control, ROS2, Sim-to-Real Transfer